Financial News Intelligence Agent

Reviewed architecture and functional specification

US + Israel • News-only financial implications • Telegram alerts

MISSION

Monitor approved news and financial sources, identify genuinely new and material events, infer evidence-backed implications for public companies and sectors, and notify only when conclusions pass explicit quality gates.

Architecture review: decisions

- **One orchestrator, fixed workflow.** A swarm of autonomous agents adds complexity without clear benefit for this sequential task.
- **Strict news-only scope.** Do not add price feeds, technical indicators, portfolio data, or sources outside the configured allowlist.
- **Separate implication from trade validation.** News alone cannot prove that a stock is mispriced or that now is a good entry point.
- **Hard gates beat confidence.** Failed verification, weak company exposure, or unresolved contradictions block an opportunity alert.
- **Persist event state.** Track events, source articles, decisions, and notifications so repeated headlines do not trigger repeated alerts.
- **Verify source access.** Confirm feed/API availability, licensing, freshness, language coverage, and permitted retrieval before production.

Out of scope

Portfolio management, trade execution, price targets, technical analysis, market-price validation, and claims of guaranteed returns. Outputs are news-derived hypotheses—not automatic buy/sell instructions.

1. Scope and processing chain

The agent's sole job is to turn new information from approved news and financial websites into concise, evidence-linked financial implications for public companies and sectors. A successful run may send no notifications.

1. Collect

Fetch headlines and available article content from approved RSS feeds, APIs or permitted pages. Record publisher, URL, publication/retrieval timestamps, language and access status.

2. Normalize & deduplicate

Cluster syndicated, repeated and follow-up reports into one event. Assign a persistent event ID; preserve corrections and meaningful updates.

3. Filter

Reject stale, irrelevant, repetitive or immaterial items. Prioritize novelty and plausible financial significance, not emotional wording or popularity.

4. Verify

Separate confirmed facts from claims and speculation. Compare approved sources; identify genuinely independent corroboration, conflicts and gaps.

5. Analyze impact

Trace event → economic mechanism → sector → company exposure → possible financial consequence → potential stock implication.

6. Challenge

Test assumptions, opposing effects, dependencies, time horizon and thesis invalidation conditions. Reject unsupported ticker associations.

7. Decide

Apply explicit verification, materiality, exposure, novelty and evidence-quality gates. Scores must not override failed hard gates.

8. Notify or log

Send Telegram only for enabled qualifying statuses. Otherwise persist the decision and reason without an urgent notification.

Example causal chain

Event	Mechanism	Exposure	Conclusion
A confirmed disruption affects oil exports	Sustained supply reduction could raise crude prices	Producers may benefit; fuel-intensive businesses may face higher costs	Conditional sector/company implication—not an automatic buy signal

Operating principles

- **Source-bounded:** use only configured sources and permitted linked content.
- **Evidence-first:** label facts, attributed claims, forecasts, opinions and unknowns separately.
- **Causal, not keyword-based:** every named ticker needs an explicit, evidence-supported connection.
- **Balanced:** include beneficiaries, losers, second-order effects and counterarguments.
- **Abstain safely:** if the evidence is weak or contradictory, do not force a conclusion.

2. Architecture and skills

Recommended architecture: one orchestrator, deterministic stage-by-stage workflow, focused LLM calls, persistent event state, and application code for thresholds, validation, deduplication and delivery. Start with Python modules and strict schemas; add multi-agent complexity only if testing proves it improves results.

Skill	Responsibility	Guardrail
extract_financial_event	Normalize event, facts, entities, timestamps and source links.	No recommendations; never guess missing facts.
verify_event	Record evidence quality, corroboration, conflicts and status.	Copied reports are not independent confirmations.

Skill	Responsibility	Guardrail
analyze_financial_impact	Build causal chains; map exposed sectors/companies, risks and horizon.	No ticker based only on keyword or sector association.
evaluate_alert	Apply hard gates for relevance, novelty, materiality and evidence.	A score cannot override a failed hard gate.
publish_notification	Format approved alerts, send Telegram, deduplicate and log delivery.	Never publish invalid or unchanged duplicate alerts.

MVP components

- **Source adapters:** one per RSS/API/approved site, with per-source polling, timeout, retry, rate-limit and freshness checks.
- **Scheduler:** EventBridge + Lambda for short bounded runs; ECS Fargate for longer or dependency-heavy processing.
- **Storage:** PostgreSQL or a simpler database for articles, event clusters, analysis versions, decisions and alert history.
- **LLM:** schema-constrained extraction and causal analysis. Treat article text as untrusted input.
- **Rules layer:** enforce allowlists, required fields, hard gates, idempotency and notification policy.
- **Delivery:** Telegram Bot API; keep credentials in a secret manager, never in source code.

NEWS-ONLY BOUNDARY

The agent may conclude that an event could benefit or harm a company based on approved news. It must not claim a stock is mispriced, identify an attractive entry point, or justify a price target without approved market-price/valuation evidence. Label outputs as news-derived hypotheses.

3. Decision logic and Telegram contract

Use statuses that accurately describe what the evidence supports. Numeric scores may help prioritize but must never be presented as probabilities of profit.

Status	Meaning	Default action
OPPORTUNITY_CANDIDATE	Verified, material event with a clear causal link to exposed companies/sectors.	Telegram only if every hard gate passes; label as a hypothesis.
WATCHLIST	Plausible thesis but evidence, a key fact or catalyst is missing.	Optional digest or later update; not urgent by default.
INFORMATIONAL	Relevant news without a sufficiently actionable implication.	Store; optional scheduled digest.
UNVERIFIED	Material claims cannot be confirmed or reports conflict.	No trade-oriented alert; log uncertainty.
NO_ACTION	No meaningful, defensible financial conclusion.	Store reason; no notification.

Every Telegram alert must include

- Event title/time, concise neutral summary, and original source links.
- Verification status; clearly separate facts from claims or forecasts.
- Potentially affected sectors and ticker/exchange mappings only where supported by evidence.
- Causal explanation, important assumptions, conditional direction and expected horizon.
- Principal counterargument, risks, unknowns, and thesis-invalidation condition.
- Status and reason for alert; explicitly state that market pricing was not checked.

Hard gates before an opportunity alert

- Event is new and not just duplicate or stale coverage.
- Material facts meet the configured verification standard; conflicts are surfaced.
- Clear causal mechanism and evidence for the named company’s actual exposure.
- No unsupported assumptions, fabricated metrics, or missing critical fields.
- Structured output and source links pass validation; no unchanged alert already exists.

Reliability and security controls

- Explicit allowlist; the model cannot add sources itself.
- Ignore instructions embedded in retrieved articles or webpages.
- Persist stable event IDs, article references, analysis versions, decision reasons and delivery state.
- Retry transient failures idempotently; track source lag, parser errors, paywalls and missing content.
- Maintain regression cases for duplicates, false positives, conflicts, weak causal links and missed events.

4. Recommended source allowlist: US + Israel

Start with a compact, complementary set. Before production, verify current feed/API access, licensing, update frequency and permitted retrieval for every source. A strong publisher is not necessarily available through a public RSS feed.

Core international financial news

Source	Use	Website
Reuters	Breaking corporate, geopolitical, commodity and macro news	www.reuters.com
Bloomberg	Financial markets, companies, commodities and macro context	www.bloomberg.com
CNBC	US markets, earnings and investor-focused coverage	www.cnbc.com
Financial Times	International economics, trade, energy and global business	www.ft.com

US primary and specialist sources

Source	Use	Website
SEC EDGAR	Official company filings and material disclosures	www.sec.gov/edgar/search
Federal Reserve	Monetary policy, official statements and speeches	www.federalreserve.gov/feeds/feeds.htm
EIA	Oil/gas data, inventories and supply-demand reports	www.eia.gov/tools/rssfeeds
BLS	Employment and inflation-related releases	www.bls.gov
BEA	GDP, income, trade and national economic accounts	www.bea.gov
US Department of Defense	Official defense announcements and procurement-related news	www.defense.gov/News
Company investor relations	Official earnings, guidance, contracts and company announcements	Relevant company IR site

Israeli business news and official disclosures

Source	Use	Website
Globes	Israeli business, markets, technology, defense, energy and macro news	www.globes.co.il
Calcalist	Israeli business, companies, technology and local economic developments	www.calcalist.co.il
TheMarker	Israeli business, finance, markets and economic policy	www.themarker.com
TASE	Official exchange information and market announcements	www.tase.co.il/en
MAYA	Israeli public-company filings and official disclosures	maya.tase.co.il

Source and ingestion policy

- Start with roughly 8–12 source groups; expand only when a source adds unique, timely, relevant coverage.
- Use RSS/supported APIs where available; use licensed feeds when required. Do not assume unrestricted scraping or redistribution rights.
- Prefer official primary sources to verify material claims. Syndicated copies are not independent corroboration.
- Support English and Hebrew. Preserve original language/headline and normalize events to one internal schema.
- Configure polling per source. Fifteen minutes is only a starting point for sources whose feed/API and rate limits support it.
- If only a headline/snippet is available, do not pretend the full article was read.

5. Implementation plan and acceptance checks

Build the smallest system that proves the workflow is reliable before adding more sources or agent complexity.

Phase 1 — Ingestion

Source configuration, per-source adapters, timestamps, retries, source health, and persistent article records.

Phase 2 — Event grouping

Deduplication, event IDs, updates/corrections, novelty detection and audit trail.

Phase 3 — Evidence and impact

Structured extraction, verification, company/sector mapping, causal chains and counterarguments.

Phase 4 — Decision and Telegram

Hard gates, status policy, message templates, idempotent delivery and secret management.

Phase 5 — Evaluation

Label good alerts, duplicates, false positives, conflicting reports and weak causal links; revise rules from results.

Metrics to monitor

- Source freshness and collection success rate.
- Duplicate-alert rate and repeated-event rate.
- Percentage of alerts with complete source citations and a clear causal chain.
- Human-reviewed false-positive rate and alerts that should have been suppressed.
- Time from publication to detection/notification.
- Which sources contribute unique relevant events versus duplicate noise.

Suggested starting configuration

Setting	Initial recommendation
Polling	Per-source configurable; about 15 minutes only where supported by feed/API limits.
Coverage	US + Israeli public companies/sectors; English and Hebrew sources.
Alerting	Telegram; OPPORTUNITY_CANDIDATE only after all hard gates pass.
Market data	Disabled. No price, volume, technical or valuation claims.
No action	Persist the decision and reason; do not notify for every run.

SUCCESS CRITERION

The agent is excellent at one job: turning new information from approved US and Israeli news/financial sources into selective, source-grounded financial implications. If no sufficiently supported conclusion exists, it sends no notification.